

Registering Our Application with LinkedIn ,X(Twitter) and working with Insta,facebook,LinkedIn,X(Twitter) APIs (Step-by-Step)

Since LinkedIn does not allow third-party applications to publish posts without registration, we must first register our application in the LinkedIn Developer Portal.

Step 1 – Create a LinkedIn Account

- 1. Open your browser.**
- 2. Visit <https://www.linkedin.com/>**
- 3. If you already have a LinkedIn account, log in.**
- 4. If you don't have one, click Join Now and create an account.**
- 5. Verify your email address.**
- 6. Complete your LinkedIn profile (recommended).**

Outcome: You now have a LinkedIn account that can access the LinkedIn Developer Portal.

Step 2 – Open the LinkedIn Developer Portal

- 1. Visit <https://www.linkedin.com/developers/>**
- 2. Click My Apps.**
- 3. Log in using your LinkedIn account if prompted.**
- 4. Click Create App.**

Outcome: LinkedIn opens the application registration form.

Step 3 – Fill in Application Details

Enter the following details:

Application Name

Example:

Social Media Scheduler

LinkedIn Page

- Every LinkedIn application must be associated with a LinkedIn Company Page.
- Create a Company Page if you don't already have one.
- Select that Company Page.

Privacy Policy URL

Example

<https://yourdomain.com/privacy>

Application Logo

- Click Upload Logo
- Upload your application logo.
- Recommended size: 100 × 100 pixels or higher.

Business Email

Enter your official email address.

Example

admin@yourcompany.com

Application Description

Example

A platform to schedule and publish content across multiple social media platforms.

Click

Create App

Step 4 – Verify Your Application

LinkedIn may ask you to verify ownership of the Company Page.

Follow the verification process if prompted.

Once verified,

the application dashboard opens.

Step 5 – Open the Authentication Tab

From the left menu,

Click

Auth

This section contains all OAuth settings.

Step 6 – Generate Client Credentials

Inside the Auth page,

LinkedIn automatically generates

Client ID

Example

86abx8v...

Copy it.

Client Secret

Click

Show Client Secret

Copy it.

Never expose it to React.

Step 7 – Configure OAuth Redirect URLs

Scroll down to

Authorized Redirect URLs

Click

Add Redirect URL

Development

<http://localhost:8000/auth/linkedin/callback>

Production

<https://yourdomain.com/auth/linkedin/callback>

Click

Save

Step 8 – Configure OAuth Scopes

Open

Products



Request access to the required products.

Typical scopes include

openid

profile

email

For posting content,

request permissions such as

w_member_social

or the relevant LinkedIn Marketing permissions available for your app type.

These permissions allow:

- **Publish posts**
- **Read profile**
- **Read email**
- **Access organization pages (if approved)**

Step 9 – Add Required Products

Open

Products

Enable products such as

- **Sign In with LinkedIn**
- **Share on LinkedIn (or the currently available posting product)**
- **Marketing Developer Platform (if your use case qualifies and is approved)**

Some products require LinkedIn approval before they become active.

Step 10 – Verify Application Status

Check

Application Status

Ensure

App Created

Redirect URI Added

Products Enabled

OAuth Configured

Step 11 – Store Credentials

Inside FastAPI

Create

LINKEDIN_CLIENT_ID=xxxxxxx

LINKEDIN_CLIENT_SECRET=xxxxxxx

LINKEDIN_REDIRECT_URI=http://localhost:8000/auth/linkedin/callback

Never expose these values to the frontend.

Step 12 – Registration Complete

You now have

- Client ID
- Client Secret
- Redirect URI
- OAuth Configuration
- Required Products Enabled

The next step will be implementing OAuth so users can connect their LinkedIn accounts.

Registering Our Application with X (Twitter)

Before publishing tweets through the API, our application must be registered in the X Developer Portal.

Step 1 – Create an X Account

1. Visit <https://x.com/>
2. Create an account if you don't already have one.
3. Verify your email address and phone number.

Outcome: You now have an X account.

Step 2 – Apply for a Developer Account

1. Visit <https://developer.x.com/>
2. Click

Developer Portal

3. Click

Apply

4. Complete the developer application.
5. Describe how you plan to use the API.

6. Submit your application.

Wait until X approves your developer access.

Step 3 – Open the Developer Portal

After approval,

Log in to

Developer Portal.

You will see

- **Projects**
- **Apps**
- **Keys and Tokens**

Step 4 – Create a Project

Click

Create Project

Enter

Project Name

Example

Social Media Scheduler

Choose the use case.

Click

Next

Step 5 – Create an App

After creating the project,

Click

Create App

Enter

Application Name

Example

Scheduler API

Click

Create

Step 6 – Enable OAuth 2.0

Open

App Settings



User Authentication Settings

Click

Set Up

Enable

OAuth 2.0

Step 7 – Configure App Permissions

Select

- **Read**
- **Write**
- **Offline Access (if refresh tokens are required)**

For publishing tweets,

Write permission is mandatory.

Step 8 – Configure Callback URL

Add

Development

<http://localhost:8000/auth/twitter/callback>

Production

<https://yourdomain.com/auth/twitter/callback>

Click

Save

Step 9 – Configure Website URL

Enter

Development

<http://localhost:3000>

Production

<https://yourdomain.com>

Click

Save

Step 10 – Configure OAuth Scopes

Select the required scopes.

Common scopes include

tweet.read

tweet.write

users.read

offline.access

media.write

These scopes allow your application to:

- **Read tweets**
- **Publish tweets**
- **Read user information**
- **Upload media**
- **Refresh expired access tokens**

Step 11 – Generate API Credentials

Navigate to

Keys and Tokens

Generate

- **Client ID**
- **Client Secret**

Copy both values.

Step 12 – Store Credentials Securely

Create

TWITTER_CLIENT_ID=xxxxxxxxx

TWITTER_CLIENT_SECRET=xxxxxxxxx

TWITTER_REDIRECT_URI=http://localhost:8000/auth/twitter/callback

Never expose these credentials in the React frontend.

Step 13 – Verify Configuration

Ensure the following are completed:

Developer Account Approved

Project Created

App Created

OAuth 2.0 Enabled

Callback URL Added

Website URL Added

Required OAuth Scopes Selected

Client ID Generated

Client Secret Generated

Step 14 – Registration Complete

Your X application is now ready for OAuth integration.

You now have:

- **Client ID**
- **Client Secret**
- **Redirect URI**
- **OAuth Configuration**
- **Required Scopes**

The next implementation step is allowing users to click "Connect X", complete the OAuth flow, obtain an Access Token, and then use the X API to publish tweets, upload media, and retrieve analytics from your FastAPI backend.

1. FACEBOOK GRAPH API IMPLEMENTATION

Feature 1 – Connect Facebook Account

Documentation to Open

Meta Graph API Documentation



Facebook Login



OAuth Documentation

Implement

- **Register App in Meta**
- **Enable Facebook Login**
- **Configure Redirect URI**
- **Generate Client ID**
- **Generate Client Secret**

OAuth Scope

pages_show_list

pages_manage_posts

pages_read_engagement

business_management

FastAPI Endpoint

GET /connect/facebook

↓

Generate OAuth URL

↓

Redirect user to Facebook

Feature 2 – Receive Authorization Code

Google is not involved here—Meta redirects back to your FastAPI callback.

GET /auth/facebook/callback

Receive

Authorization Code

Exchange it for

- **Access Token**
- **Long-lived Access Token**

Store

- **User ID**
- **Page ID**

- Access Token
- Expiry

Feature 3 – Get User Pages

Documentation

Pages API

Endpoint

GET /me/accounts

Include

Authorization Header

Bearer Access Token

Store

- Page ID
- Page Name
- Category

Display Pages in React.

Feature 4 – Publish Facebook Post

Documentation

Page Feed API

Endpoint

POST /{page-id}/feed

Include

Headers

Authorization Bearer Token

Body

- Message
- Image URL (optional)

- Link
- Scheduled Publish Time

Response

Store

- Facebook Post ID
- Status
- Publish Time

Update dashboard.

Feature 5 – Read Facebook Analytics

Documentation

Page Insights API

Retrieve

- Likes
- Reach
- Impressions
- Comments
- Shares
- Engagement

Display graphs in React.

2. INSTAGRAM GRAPH API IMPLEMENTATION

Feature 1 – Connect Instagram

Documentation

Instagram Graph API

Register App



Enable Instagram Graph API



Configure Redirect URI



OAuth



Access Token

Store

- **Instagram User ID**
- **Access Token**

Feature 2 – Retrieve Instagram Business Account

Endpoint

GET /me/accounts



Retrieve Facebook Page



Retrieve

Instagram Business Account ID

Store

Instagram User ID

Feature 3 – Create Media Container

Instagram does NOT publish directly.

First create

Media Container.

Documentation

Media Endpoint

Endpoint

POST /{ig-user-id}/media

Body

- Image URL
- Caption

Headers

Bearer Token

Response

Creation ID

Store

Creation ID

Feature 4 – Publish Media

Documentation

Media Publish Endpoint

Endpoint

POST /{ig-user-id}/media_publish

Body

creation_id

Response

Instagram Media ID

Store

- Media ID
- Publish Time

Update Dashboard.

Feature 5 – Analytics

Endpoint

Insights API

Retrieve

- **Reach**
- **Impressions**
- **Likes**
- **Saves**
- **Comments**
- **Followers**
- **Profile Visits**

Display charts.

3. LINKEDIN API IMPLEMENTATION

Feature 1 – Connect LinkedIn

Open

LinkedIn Developer Portal



Create App



OAuth



Redirect URI



Client ID



Client Secret



Access Token

Store

- **Member ID**

- Access Token

Feature 2 – Retrieve Profile

Documentation

People API

Endpoint

GET /me

Retrieve

- Name
- Profile Picture
- Member ID

Display connected account.

Feature 3 – Publish Post

Documentation

Posts API

Endpoint

POST /posts

Headers

Bearer Token

Content-Type

application/json

Body

- Author ID
- Text Content
- Visibility
- Media (optional)

Response

LinkedIn Post ID

Store

- **Post ID**
- **Publish Time**

Feature 4 – Upload Images

Documentation

Assets API

Flow

Register Upload

↓

Receive Upload URL

↓

Upload Image

↓

Receive Asset ID

↓

Use Asset ID

inside

Posts API

Feature 5 – Analytics

Documentation

Organization Analytics API

Retrieve

- **Likes**
- **Comments**
- **Shares**

- Followers
- Engagement

Store in PostgreSQL.

4. X (TWITTER) API IMPLEMENTATION

Feature 1 – Connect X

Open

X Developer Portal



Create App



OAuth 2.0



Redirect URI



Client ID



Client Secret



Access Token

Store

- User ID
- Access Token

Feature 2 – Retrieve User

Endpoint

GET /2/users/me

Retrieve

- Username
- Display Name
- Profile Image

Display Connected Account.

Feature 3 – Publish Tweet

Documentation

Create Tweet Endpoint

Endpoint

POST /2/tweets

Headers

Bearer Token

Body

```
{  
  "text": "Hello World"  
}
```

Response

Tweet ID

Store

- Tweet ID
- Publish Time

Feature 4 – Upload Images

Images cannot be uploaded directly with Create Tweet.

First

Use

Media Upload API



Receive

Media ID



Pass

Media ID

inside

Create Tweet API.

Feature 5 – Analytics

Retrieve

- **Likes**
- **Replies**
- **Retweets**
- **Quotes**
- **Impressions**

Display Dashboard.